

Existence and finite-time stability of discrete fractional-order complex-valued neural networks with time delays

Xingxing You^a, Qiankun Song^{b,c,*}, Zhenjiang Zhao^d

^a School of Economic and Management, Chongqing Jiaotong University, Chongqing 400074, China

^b Department of Mathematics, Chongqing Jiaotong University, Chongqing 400074, China

^c State Key Laboratory of Mountain Bridge and Tunnel Engineering, Chongqing Jiaotong University, Chongqing 400074, China

^d Department of Mathematics, Huzhou University, Huzhou 313000, China

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ABSTRACT

Without decomposing complex-valued systems into real-valued systems, the existence and finite-time stability for discrete fractional-order complex-valued neural networks with time delays are discussed in this paper. First of all, in order to obtain the main results, a new discrete Caputo fractional difference equation is proposed in complex field based on the theory of discrete fractional calculus, which generalizes the fractional-order neural networks in the real domain. Additionally, by utilizing Arzela–Ascoli's theorem, inequality scaling skills and fixed point theorem, some sufficient criteria of delay-dependent are deduced to ensure the existence and finite-time stability of solutions for proposed networks. Finally, the validity and feasibility of the derived theoretical results are indicated by two numerical examples with simulations. Furthermore, we have drawn the following facts: with the lower order, the discrete fractional-order complex-valued neural networks will achieve the finite-time stability more easily.

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1. Introduction

Fractional-order calculus, as the extension of integer-order calculus, was initially established by Leibniz in 1695. There are many masters of mathematics such as Liouville, Riemann, Abel, Fourier, who have spent many years on developing and perfecting its theory, constantly (Podlubny, 1999). As the deepening of the theoretical research, fractional-order calculus has been diffusely applied in science and engineering (Park, 2006; Park, Kwon, & Lee, 2008), such as signal processing, control systems and associative memory. Compared with classical integer-order systems, the unique advantage of fractional-order system is that it offers an excellent instrument for the description of memory and hereditary properties of various materials and processes (Kwon, Park, Park, Lee, & Cha, 2014; Lenka & Banerjee, 2016; Stamova, 2014b; Thanh, Trinh, & Phat, 2017; Trigeassou, Maamri, Sabatier, & Oustaloup, 2011). Based on this feature, fractional-order cellular neural networks were first proposed by Arena, Caponetto, Fortuna, and Porto (1998). However, when the fractional-order neural networks are used in hardware implementation of the system, time delay is frequently occurred and has a detrimental

effect on the quality of the control system (Fan, Huang, Li, Xia, & Chen, 2018; Jia, Huang, Li, Cao, & Alsaedi, 2019; Wu & Zeng, 2014). Because of the existence of delays, the disturbance cannot be detected early, and the control effect cannot be effective in time, resulting in instability, poor control or even out of control (Arik, 2016, 2019; Liu, 2016; Song, Yang, Li, Huang, & Chen, 2017). Hence, people are committed to researching the stability of fractional-order neural networks with time delays and have also obtained many important and interesting theoretical results. For example, the scholars have established some stability conditions for fractional-order real-valued neural networks (RVNNs) without or with various types of delays (Ji, He, Wu, & Zhang, 2015; Jia et al., 2019; Stamova, 2014a; Wang, Yu, Wen, Zhang, & Yu, 2015; Wu, Liu, Huang, & Zeng, 2017; Yang, He, Wang and Wu, 2018; Yu, Hu, & Jiang, 2012; Zhang et al., 2018; Zhang, Yu and Yu, 2017). As is well known, RVNNs perform less well in detection of symmetry problem, two-dimensional (2D) affine transformations and the exclusion OR (XOR) problem (Yang, Yu, Hu and Jiang, 2018). Compared with RVNNs, complex-valued neural networks (CVNNs), whose state vectors, connection weights, activation functions, external inputs are complex-valued, have much more complicated properties (Wang, Wang and Shen, 2019). Hence, it can directly deal with the issues which cannot be solved by RVNNs. More and more stability criteria for fractional-order CVNNs with several types of delays were considered (Rakkiyappan, Cao, & Velmurugan, 2015; Wang, Song, Liu, Zhao, & Alsaadi, 2017b; Zhang, Song and Zhao, 2017).

* Corresponding author at: Department of Mathematics, Chongqing Jiaotong University, Chongqing 400074, China.

E-mail addresses: youxingxing11@163.com (X. You), qiankunsong@163.com (Q. Song), zhaozjcn@163.com (Z. Zhao).

In recent decades, most of researches on stability are referred to stability in Lyapunov sense, such as asymptotic, exponential and uniform stability (Fan, Huang, Shen, & Cao, 2019; Peng, Zeng, & Wang, 2017; Wang, Chu and Zhang, 2019; Yang, Chu, & Zhang, 2019). However, these results about stability mentioned above pay attention to the state of the system when the time interval tends to infinity. Since most actual networks spend finite time intervals in processing information, the main concern is the finite-time behavior in some practical applications. Therefore, the theory of finite-time stability was put forward and introduced by Kamenkov (1953). Compared with Lyapunov stability, the finite-time stability not only gains the estimated time of system, but also handles system with prescribed bounds and finite time intervals. The issues of finite-time stability for fractional-order system have invoked increasing interest of scholars from kinds of fields, an increasing number of theories have been proposed. For example, see Chen, Liu, Wu, He, and Chai (2016), Ran Chao, Xin Dong, and Li Ping (2013), Wu, Lu, and Chen (2015) and Yang, Song, Liu, and Zhao (2015) for the fractional-order RVNNs, Ding, Cao, Zhao, and Alsaadi (2017), Rakkiyappan, Velmurugan, and Cao (2014) and Wang, Song, Liu, Zhao, and Alsaadi (2017a) for the fractional-order CVNNs.

It should be noted that all of the above mentioned results focus on continuous-time systems. However, in real world applications, the continuous-time and discrete-time systems are two complementary aspects, and thus, a nature question arises: whether the discrete-time systems possess the similar dynamical behaviors to their continuous-time counterparts? In theoretical research, not all discrete operators have the same properties as continuous operators, so it is imperative to research the dynamic behavior of discrete fractional calculus. In particular, the existence analysis and stability criteria founded for continuous-time CVNNs are unsuitable for discrete-time systems sometimes. As is known to all, researchers often consider continuous-time system discretization to analyze its dynamic behavior in computer simulation and analysis. Furthermore, when the control rules are implemented digitally, the discrete-time system becomes more important than the continuous-time system. Discrete fractional calculus can describe better the dynamics of complex discrete systems with the long-range interaction traits. Therefore, researching dynamical properties of discrete-time fractional-order CVNNs is open and challenging. Recently, some significant results of discrete fractional differential equations have been reported (Abdeljawad, 2011, 2013; Alzabut, Abdeljawad, & Baleanu, 2018; Atici & Eloe, 2009, 2012; Ganji & Gharari, 2018; Goodrich & Peterson, 2015; Wu & Baleanu, 2014). In Atici and Eloe (2009, 2012), the authors discussed an initial value problem for two types of discrete fractional equation and sum equation. In Abdeljawad (2011, 2013), two types of fractional differences for Riemann and Caputo were introduced. In Wu and Baleanu (2014), the authors put forward a new formidable tool to investigate discrete fractional logistic map and analyzed its chaotic behaviors. In Alzabut et al. (2018), the authors researched the existence and uniqueness for non-linear discrete fractional difference equations by using discrete Arzela–Ascoli’s theorem, inequality scaling skills and fixed point theorem. Nevertheless, to our knowledge, there are very few results on discrete fractional-order CVNNs with time delays. This motivates our present research.

Considering the above analysis, this paper is dedicated to researching the issues of existence and finite-time stability for discrete fractional-order CVNNs with time delays. The main contributions of the paper are highlighted as follows.

- (i) As far as we know, this paper is the first attempt to address the discrete fractional-order neural networks in complex field. Two new lemmas in complex field are established under discrete fractional-order CVNNs.

- (ii) This paper directly investigates the existence and finite-time stability for discrete fractional-order CVNNs by employing the Krasnoselskii fixed point theorem and stability theory in complex field, instead of decomposing complex-valued systems into real-valued systems, which is different from the existing literature Ding et al. (2017), Rakkiyappan et al. (2014) and Wang et al. (2017a).
- (iii) By analyzing the conditions in the theorems, we come to the following conclusions: with the lower order, the discrete fractional-order CVNNs will achieve the finite-time stability more easily.

The organizational structure of this paper is shown below. Some definitions and assumptions of discrete fractional-order CVNNs are introduced in Section 2. At the same time, some necessary lemmas are also added. Section 3 presents the sufficient criteria of delay-dependent for existence and finite-time stability of solutions by utilizing Arzela–Ascoli’s theorem, inequality scaling skills and fixed point theorem. In Section 4, two numerical examples which manifest the validity and feasibility of the acquired results are provided. Finally, the conclusion of this paper is provided in Section 5.

Notations: Throughout this paper, we adopt discrete Caputo fractional derivative, ${}^c\nabla_a^\alpha$ stands for a class of Caputo fractional difference operator with the order α ($0 < \alpha < 1$). For real number a , $\mathbb{N}_a = \{a, a+1, a+2, \dots\}$. Let $\mathbb{R}^n$ and $\mathbb{C}^n$ denote the set of all n -dimensional real and complex vectors, $\mathbb{R}^{n \times m}$ and $\mathbb{C}^{n \times m}$, respectively, stand for the set of $n \times m$ real and complex matrices. Let $h = x + yi$ be a complex number, where $x, y \in \mathbb{R}$ are the real and imaginary parts of h , $i = \sqrt{-1}$ denotes the imaginary unit. h^* is the conjugate of h , $|h| = \sqrt{h^*h}$ represents the module of h . For positive constants P, Q , Gamma function and Beta function are defined by $\Gamma(P) = \int_0^{+\infty} t^{P-1}e^{-t}dt$ and $B(P, Q) = \int_0^1 t^{P-1}(1-t)^{Q-1}dt$, respectively. Let Banach space of continuous n -complex vector bounded functions (sequences) which defined on $[-\tau + a, a]$ be expressed by $L([-\tau + a, a], \mathbb{C}^n)$, whose norm is denoted by $\|\phi\|_C = \sup_{-\tau+a \leq t \leq a} \sum_{j=1}^n |\phi_j(t)|$ for $\phi \in L([-\tau + a, a], \mathbb{C}^n)$. Furthermore, for $h(t) = (h_1(t), h_2(t), \dots, h_n(t))^T \in L([a, T], \mathbb{C}^n)$, $\|h(t)\| = \sup_{a \leq t \leq T} \sum_{j=1}^n |h_j(t)|$ represents the norm of $h(t)$, where $T \in \mathbb{N}_a$. The set $\mathcal{U}$ is defined by $\mathcal{U} = \{h : h = (h_1, h_2, \dots, h_n)^T, \|h\| \leq \eta, h_j \in L([a, T], \mathbb{C})\}$.

2. Model description and preliminaries

We consider a class of discrete fractional-order CVNNs with time delays in this paper,

$${}^c\nabla_a^\alpha h_j(t) = -c_j h_j(t) + \sum_{k=1}^n a_{jk} f_k(h_k(t)) + \sum_{k=1}^n b_{jk} g_k(h_k(t - \tau)) + W_j, \quad j = 1, 2, \dots, n, \quad t \in \mathbb{N}_a, \quad (1)$$

or other equivalent vector expression

$${}^c\nabla_a^\alpha h(t) = -Ch(t) + Af(h(t)) + Bg(h(t - \tau)) + W, \quad t \in \mathbb{N}_a, \quad (2)$$

where ${}^c\nabla_a^\alpha$ denotes the Caputo fractional difference operator (starting from a) with order α ($0 < \alpha < 1$); $h(t) = (h_1(t), h_2(t), \dots, h_n(t))^T \in \mathbb{C}^n$ is complex state vector; $C = \text{diag}(c_1, c_2, \dots, c_n) \in \mathbb{R}^{n \times n}$ is self-feedback connection weight matrix and $c_j > 0$; $A = (a_{jk})_{n \times n} \in \mathbb{C}^{n \times n}$ and $B = (b_{jk})_{n \times n} \in \mathbb{C}^{n \times n}$ are connection weight matrices; $f(h(t)) = (f_1(h_1(t)), f_2(h_2(t)), \dots, f_n(h_n(t)))^T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ and $g(h(t - \tau)) = (g_1(h_1(t - \tau)), g_2(h_2(t - \tau)), \dots, g_n(h_n(t - \tau)))^T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ are vector-valued activation functions; Positive integer τ denotes time delay; $W = (W_1, W_2, \dots, W_n)^T \in \mathbb{C}^n$ denotes the external input of system (1). The initial condition of system (1) is given by

$$h_j(s) = \phi_j(s), \quad s \in [-\tau + a, -\tau + a + 1, \dots, a], \quad (3)$$

where $\phi_j(s) \in L([-\tau + a, a], \mathbb{C})$.

To analyze existence and finite-time stability of solutions of CVNNs (1), the following assumptions on activation functions are necessary.

Assumption 1. There exist positive constants l_k, λ_k , such that

$$|f_k(\alpha_1) - f_k(\alpha_2)| \leq l_k |\alpha_1 - \alpha_2|, \quad k = 1, 2, \dots, n,$$

$$|g_k(\alpha_1) - g_k(\alpha_2)| \leq \lambda_k |\alpha_1 - \alpha_2|, \quad k = 1, 2, \dots, n,$$

for any $\alpha_1, \alpha_2 \in \mathbb{C}$.

Assumption 2. There exist two positive constants M and N such that $|f_k(\alpha)| \leq M$ and $|g_k(\alpha)| \leq N$ for any $\alpha \in \mathbb{C}$.

Definition 1 (Ganji & Gharari, 2018; Goodrich & Peterson, 2015).

(i) For a natural number e and $t \in \mathbb{R}$, the e ascending function of t can be defined by

$$t^{\bar{e}} = \prod_{k=0}^{e-1} (t + k), \quad t^{\bar{0}} = 1.$$

(ii) For $\alpha \in \mathbb{R}$, the α ascending function can be defined by

$$t^{\bar{\alpha}} = \frac{\Gamma(t + \alpha)}{\Gamma(t)}, \quad t \in \mathbb{R} - \{\dots, -2, -1, 0\}, \quad 0^{\bar{\alpha}} = 0.$$

In addition, we define iteratively operator $\nabla^e = \nabla(\nabla^{(e-1)})$.

Definition 2 (Alzabut et al., 2018; Zhou, 2007). A set $\bar{U}$ of sequences in Banach space L is uniformly Cauchy if for any $\mu > 0$, there exists integer N such that $|\bar{h}(t_1) - \bar{h}(t_2)| < \mu$ whenever $t_1, t_2 > N$ for every $\bar{h} = \{\bar{h}(s)\} \in \bar{U}$.

Definition 3 (Wang et al., 2017a). Set a time $T > 0$, given positive constants δ, ε and $\delta < \varepsilon$. A solution $\bar{h}(t, t_0, \phi)$ of system (1) is finite-time stable with respect to $(t_0, \delta, \varepsilon, \Theta)$, if for any solution $u(t, t_0, \psi)$, $\|\phi - \psi\|_{\mathbb{C}} < \delta$ implies $\|\bar{h}(t, t_0, \phi) - u(t, t_0, \psi)\| < \varepsilon, \forall t \in \Theta = [t_0, t_0 + T]$, where $\phi(t), \psi(t)$ are two initial functions of system (1) for $t \in [t_0 - \tau, t_0]$. System (1) is said to be finite-time stable with respect to $(t_0, \delta, \varepsilon, \Theta)$, if any solution $\bar{h}(t)$ of system (1) is finite-time stable with respect to $(t_0, \delta, \varepsilon, \Theta)$.

From Abdeljawad (2013) and Alzabut et al. (2018), we give the following definition in complex field.

Definition 4. For $\alpha > 0$ and real number a , let $n = [\alpha] + 1, \bar{h} : \mathbb{N}_a \rightarrow \mathbb{C}, \rho(s) = s - 1$, where $[\alpha]$ denotes the greatest integer less than α .

(i) The backward difference operator of $\bar{h}$ is defined by

$$\nabla \bar{h}(t) = \bar{h}(t) - \bar{h}(t - 1), \quad t \in \mathbb{N}_1.$$

(ii) The nabla α -order sum operator is defined by

$$\nabla_a^{-\alpha} \bar{h}(t) = \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \bar{h}(s), \quad t \in \mathbb{N}_{a+1}.$$

(iii) The nabla α -order Caputo fractional difference operator is defined by

$$\begin{aligned} {}^c \nabla_a^{-\alpha} \bar{h}(t) &= \nabla_a^{-(n-\alpha)} \nabla^n \bar{h}(t) \\ &= \frac{1}{\Gamma(n-\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{n-\alpha-1}} \nabla^n \bar{h}(s), \quad t \in \mathbb{N}_{a+1}. \end{aligned}$$

(iv) Let $\mu > -1$, then

$$\nabla_a^{-\alpha} (t - a)^{\bar{\mu}} = \frac{\Gamma(\mu + 1)}{\Gamma(\alpha + \mu + 1)} (t - a)^{\overline{\alpha+\mu}}, \quad t \in \mathbb{N}_a.$$

In particular, when $0 < \alpha < 1$ and $a = 0$, then $\mathbb{N}_0 = \{0, 1, 2, \dots\}, n = 1$. According to Definition 4, we have

$$\nabla_0^{-\alpha} \bar{h}(t) = \frac{1}{\Gamma(\alpha)} \sum_{s=1}^t (t - \rho(s))^{\overline{\alpha-1}} \bar{h}(s), \quad t \in \mathbb{N}_1,$$

$$\begin{aligned} {}^c \nabla_0^{-\alpha} \bar{h}(t) &= \nabla_0^{-(1-\alpha)} \nabla \bar{h}(t) = \frac{1}{\Gamma(1-\alpha)} \sum_{s=1}^t (t - \rho(s))^{\overline{-\alpha}} \nabla \bar{h}(s), \\ t &\in \mathbb{N}_1, \end{aligned}$$

$$\nabla_0^{-\alpha} t^{\bar{\mu}} = \frac{\Gamma(\mu + 1)}{\Gamma(\alpha + \mu + 1)} t^{\overline{\alpha+\mu}}, \quad t \in \mathbb{N}_0.$$

Remark 1. According to the above definition, we can get $\nabla_a^{-\alpha} (uf(t) + vg(t)) = u \nabla_a^{-\alpha} f(t) + v \nabla_a^{-\alpha} g(t)$ for $u, v \in \mathbb{R}, f(t), g(t) \in \mathbb{C}$.

Now, we will give the following important lemmas to prove the main results.

Lemma 1 (Abdeljawad, 2013). Assume that $\alpha > 0$ and $y : \mathbb{N}_a \rightarrow \mathbb{R}$. Then,

$$\nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} y(t) = y(t) - \sum_{k=0}^{n-1} \frac{(t-a)^{\bar{k}}}{k!} \nabla^k y(a).$$

If the order α satisfies condition $0 < \alpha \leq 1$, then

$$\nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} y(t) = y(t) - y(a).$$

We can prove that Lemma 1 holds in complex field.

Lemma 2. Assume that $\alpha > 0$ and $\bar{h} : \mathbb{N}_a \rightarrow \mathbb{C}$. Then,

$$\nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} \bar{h}(t) = \bar{h}(t) - \sum_{k=0}^{n-1} \frac{(t-a)^{\bar{k}}}{k!} \nabla^k \bar{h}(a).$$

If the order α satisfies condition $0 < \alpha \leq 1$, then

$$\nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} \bar{h}(t) = \bar{h}(t) - \bar{h}(a).$$

Proof. Let $\bar{h}(t) = x(t) + y(t)i$, where $x(t), y(t) \in \mathbb{R}$. From Lemma 1, we can get

$$\begin{aligned} \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} \bar{h}(t) &= \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} (x(t) + y(t)i) \\ &= \nabla_a^{-\alpha} ({}^c \nabla_a^{\alpha} x(t) + {}^c \nabla_a^{\alpha} y(t)i) \\ &= \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} x(t) + \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} y(t)i \\ &= x(t) - \sum_{k=0}^{n-1} \frac{(t-a)^{\bar{k}}}{k!} \nabla^k x(a) \\ &\quad + \left(y(t) - \sum_{k=0}^{n-1} \frac{(t-a)^{\bar{k}}}{k!} \nabla^k y(a) \right) i \\ &= x(t) + y(t)i - \sum_{k=0}^{n-1} \frac{(t-a)^{\bar{k}}}{k!} \nabla^k (x(a) + y(a)i) \\ &= \bar{h}(t) - \sum_{k=0}^{n-1} \frac{(t-a)^{\bar{k}}}{k!} \nabla^k \bar{h}(a). \end{aligned}$$

When the condition $0 < \alpha \leq 1$ holds, then

$$\begin{aligned} \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} \bar{h}(t) &= \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} (x(t) + y(t)i) \\ &= \nabla_a^{-\alpha} ({}^c \nabla_a^{\alpha} x(t) + {}^c \nabla_a^{\alpha} y(t)i) \\ &= \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} x(t) + \nabla_a^{-\alpha} {}^c \nabla_a^{\alpha} y(t)i \\ &= x(t) - x(a) + (y(t) - y(a))i = \bar{h}(t) - \bar{h}(a). \end{aligned}$$

The proof is completed. $\square$

Lemma 3. Consider the Caputo fractional difference system with delay:

$$\begin{cases} {}^c\nabla_a^\alpha \omega(t) = f(t, \omega(t), \omega(t - \tau)), & t \in \mathbb{N}_a, \\ \omega(s) = \phi(s), & s \in [-\tau + a, a], \end{cases} \quad (4)$$

where $0 < \alpha < 1$, $\tau > 0$, $f : \mathbb{N}_a \times \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$. Then, $\omega(t)$ is a solution of system (4) if and only if it can be written in the following form

$$\omega(t) = \phi(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} f(s, \omega(s), \omega(s - \tau)), \quad t \in \mathbb{N}_a,$$

and $\omega(s) = \phi(s)$, $s \in [-\tau + a, -\tau + a + 1, \dots, a]$.

Proof. According to Definition 4 and Lemma 2, one can get

$$\begin{aligned} \nabla_a^{-\alpha} {}^c\nabla_a^\alpha \omega(t) &= \nabla_a^{-\alpha} f(s, \omega(s), \omega(s - \tau)) \\ &= \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} f(s, \omega(s), \omega(s - \tau)) \\ &= \omega(t) - \omega(a) = \omega(t) - \phi(a). \end{aligned}$$

The proof is completed. $\square$

Lemma 4 (Zhou, 2007 Arzela–Ascoli’s Theorem). A bounded, uniformly Cauchy subset $\mathcal{U}$ of Banach space L is relatively compact.

Lemma 5 (Burton, 1998 Krasnoselskii Fixed Point Theorem). Let $\mathcal{U}$ be a nonempty, closed, convex and bounded subset of Banach space L . Suppose that A, B are two operators, such that

- (i) A is a contraction;
- (ii) for any $x, y \in \mathcal{U}$, $Ax + By \in \mathcal{U}$;
- (iii) B is continuous and $B(\mathcal{U})$ is relatively compact.

Then for $x \in \mathcal{U}$, the operator equation $Ax + Bx = x$ has a solution.

3. Main results

3.1. Existence of solutions

Theorem 1. Under Assumptions 1 and 2, if the following conditions hold

$$\mathfrak{S} < 1, \quad (5)$$

$$\frac{\wp \|\phi\|_C + \mathfrak{N}}{1 - \mathfrak{S}} \leq \eta, \quad (6)$$

then the system (1) has a solution in $\mathcal{U} = \{\mathbf{h} : \mathbf{h} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n)^T, \|\mathbf{h}\| \leq \eta, \mathbf{h}_j \in L([a, T], \mathbb{C})\}$, where

$$\begin{aligned} \mathfrak{S} &= \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \\ &\quad + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)}, \\ \wp &= 1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)}, \\ \mathfrak{N} &= \frac{1}{\alpha B(T - a, \alpha)} \|W\| + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M \\ &\quad + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N. \end{aligned}$$

Proof. Define two operators $\Phi = (\Phi_1, \Phi_2, \dots, \Phi_n)^T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ and $\Psi = (\Psi_1, \Psi_2, \dots, \Psi_n)^T : \mathbb{C}^n \rightarrow \mathbb{C}^n$, by

$$\Phi_j \mathbf{h}_j(t) = \phi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} (-c_j \mathbf{h}_j(s) + W_j)$$

and

$$\begin{aligned} \Psi_j \mathbf{h}_j(t) &= \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \left(\sum_{k=1}^n a_{jk} f_k(\mathbf{h}_k(s)) \right. \\ &\quad \left. + \sum_{k=1}^n b_{jk} g_k(\mathbf{h}_k(s - \tau)) \right), \quad j = 1, 2, \dots, n, \end{aligned}$$

where $\mathbf{h}_j(t)$ is the solution of system (1) with the initial conditions (3). The fixed point of operator is denoted by $\mathcal{E} \mathbf{h}(t) = \Phi \mathbf{h}(t) + \Psi \mathbf{h}(t)$.

When $t = a$, the operator equation $\mathbf{h}(t) = \mathcal{E} \mathbf{h}(t) = \Phi \mathbf{h}(t) + \Psi \mathbf{h}(t)$, taking the norm of $\mathbf{h}(t)$, we can get

$$\begin{aligned} \|\mathbf{h}(t)\| &= \|\Phi \mathbf{h}(t) + \Psi \mathbf{h}(t)\| = \sup_{-\tau + a \leq t \leq a} \sum_{j=1}^n |\phi_j(a)| \\ &= \sum_{j=1}^n |\phi_j(a)| = \|\phi(a)\|_C. \end{aligned} \quad (7)$$

Therefore, $\mathbf{h}(t) = \phi(a)$ is a solution in $\mathcal{U}$.

When $t \in (a, T]$, we will prove this theorem in four steps.

Step 1. We prove that operator Φ is the contraction mapping.

For $\mathbf{h}(t)$, $\mathbf{u}(t) \in \mathcal{U}$, taking the norm of $\Phi \mathbf{h}(t) - \Phi \mathbf{u}(t)$, it yields

$$\begin{aligned} &\|\Phi \mathbf{h}(t) - \Phi \mathbf{u}(t)\| \\ &= \sup_{a < t \leq T} \sum_{j=1}^n \left| \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} (-c_j (\mathbf{h}_j(s) - \mathbf{u}_j(s))) \right| \\ &\leq \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} c_j |\mathbf{h}_j(s) - \mathbf{u}_j(s)| \\ &\leq \sup_{a < t \leq T} \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \max_{1 \leq j \leq n} \{c_j\} \sum_{j=1}^n |\mathbf{h}_j(s) - \mathbf{u}_j(s)|. \end{aligned} \quad (8)$$

According to Definitions 1, 4 and $(t - a)^{\bar{0}} = 1$, it can deduce

$$\sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} (t - a)^{\bar{0}} = \Gamma(\alpha) \nabla_0^{-\alpha} (t - a)^{\bar{0}} = \frac{\Gamma(t - a + \alpha)}{\alpha \Gamma(t - a)}.$$

Therefore, (8) becomes

$$\|\Phi \mathbf{h}(t) - \Phi \mathbf{u}(t)\| \leq \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} \|\mathbf{h} - \mathbf{u}\|,$$

where the $B(T - a, \alpha)$ is Beta function, whose value relies on T, a and the order α . According to the condition $\mathfrak{S} < 1$, one can see that $\frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} < 1$. It can be concluded that the operator Φ is a contraction mapping.

Step 2. We prove that $\Phi \mathbf{u}(t) + \Psi \mathbf{h}(t) \in \mathcal{U}$ for $\mathbf{u}(t), \mathbf{h}(t) \in \mathcal{U}$.

By employing Minkowski inequality, we have

$$\begin{aligned} &\|\Phi \mathbf{u}(t) + \Psi \mathbf{h}(t)\| \\ &= \sup_{a < t \leq T} \sum_{j=1}^n \left| \phi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \right. \\ &\quad \times \left(-c_j \mathbf{u}_j(s) + W_j + \sum_{k=1}^n a_{jk} f_k(\mathbf{h}_k(s)) + \sum_{k=1}^n b_{jk} g_k(\mathbf{h}_k(s - \tau)) \right) \Big| \\ &= \sup_{a < t \leq T} \sum_{j=1}^n \left| \phi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \right. \end{aligned}$$

$$\begin{aligned}
& \times \left(-c_j u_j(s) + W_j + \sum_{k=1}^n a_{jk} f_k(\mathbf{h}_k(s)) - \sum_{k=1}^n a_{jk} f_k(0) \right. \\
& + \sum_{k=1}^n b_{jk} g_k(\mathbf{h}_k(s - \tau)) - \sum_{k=1}^n b_{jk} g_k(0) + \sum_{k=1}^n a_{jk} f_k(0) \\
& \left. + \sum_{k=1}^n b_{jk} g_k(0) \right) \Bigg| \\
& = \sup_{a < t \leq T} \sum_{j=1}^n \left| \phi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \right. \\
& \times \left(-c_j u_j(s) + W_j + \sum_{k=1}^n a_{jk} f_k(\mathbf{h}_k(s)) - f_k(0) \right) \\
& + \sum_{k=1}^n b_{jk} (g_k(\mathbf{h}_k(s - \tau)) - g_k(0)) + \sum_{k=1}^n a_{jk} f_k(0) + \sum_{k=1}^n b_{jk} g_k(0) \Bigg| \\
& \leq \sum_{j=1}^n |\phi_j(a)| + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} |W_j| \\
& + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} c_j |u_j(s)| \\
& + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |a_{jk}| |f_k(\mathbf{h}_k(s)) - f_k(0)| \\
& + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \\
& \times \sum_{k=1}^n |b_{jk}| |g_k(\mathbf{h}_k(s - \tau)) - g_k(0)| \\
& + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |a_{jk}| |f_k(0)| \\
& + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |b_{jk}| |g_k(0)|. \quad (9)
\end{aligned}$$

On the basis of Definition 1, Assumption 1 and Definition 4, it can deduce

$$\begin{aligned}
& \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |a_{jk}| |f_k(\mathbf{h}_k(s)) - f_k(0)| \\
& \leq \sup_{a < t \leq T} \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\} \sum_{k=1}^n |\mathbf{h}_k(s) - 0| \\
& \leq \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \|\mathbf{h}\|. \quad (10)
\end{aligned}$$

Similarly,

$$\begin{aligned}
& \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \\
& \times \sum_{k=1}^n |b_{jk}| |g_k(\mathbf{h}_k(s - \tau)) - g_k(0)| \\
& \leq \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |b_{jk}| |g_k(\mathbf{h}_k(s - \tau)) - g_k(0)| \\
& \leq \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a}^{\tau+a} (t - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |b_{jk}| |g_k(\mathbf{h}_k(s - \tau)) - g_k(0)|
\end{aligned}$$

$$\begin{aligned}
& + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=\tau+a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \\
& \times \sum_{k=1}^n |b_{jk}| |g_k(\mathbf{h}_k(s - \tau)) - g_k(0)| \\
& \leq \sup_{a < t \leq T} \frac{1}{\Gamma(\alpha)} \sum_{s=a}^{\tau+a} (t - \rho(s))^{\overline{\alpha-1}} \sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\} \sum_{k=1}^n |\mathbf{h}_k(s - \tau)| \\
& + \sup_{a < t \leq T} \frac{1}{\Gamma(\alpha)} \sum_{s=\tau+a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \\
& \times \sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\} \sum_{k=1}^n |\mathbf{h}_k(s - \tau)| \\
& \leq \sup_{a < t \leq T} \frac{\Gamma(t - a + 1 + \alpha)}{\alpha \Gamma(\alpha) \Gamma(t - a + 1)} \sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\} \|\phi\|_C \\
& + \sup_{a < t \leq T} \frac{\Gamma(t - a - \tau + \alpha)}{\alpha \Gamma(\alpha) \Gamma(t - a - \tau)} \sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\} \|\mathbf{h}\| \\
& \leq \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \|\phi\|_C + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \|\mathbf{h}\|, \quad (11)
\end{aligned}$$

where $T > \tau + a$. Substituting (10) and (11) in (9) and employing Assumption 2, we get

$$\begin{aligned}
& \|\Phi u(t) + \Psi \mathbf{h}(t)\| \\
& \leq \|\phi\|_C + \frac{1}{\alpha B(T - a, \alpha)} \|W\| + \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} \|u\| + \frac{1}{\alpha B(T - a, \alpha)} \\
& \times \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N \\
& + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \|\mathbf{h}\| + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \|\phi\|_C \\
& + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \|\mathbf{h}\| \\
& = \left(1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \right) \|\phi\|_C + \frac{1}{\alpha B(T - a, \alpha)} \|W\| \\
& + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M \\
& + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N + \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} \|u\| \\
& + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \|\mathbf{h}\| + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \|\mathbf{h}\| \\
& \leq \left(1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \right) \|\phi\|_C + \frac{1}{\alpha B(T - a, \alpha)} \|W\| \\
& + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M \\
& + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N + \left(\frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} \right. \\
& \left. + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \right) \eta \\
& = \mathfrak{D} \|\phi\|_C + \mathfrak{R} + \mathfrak{S} \eta. \quad (12)
\end{aligned}$$

Under the conditions (6), one obtains $\|\Phi u(t) + \Psi h(t)\| \leq \eta$, which can conclude that $\Phi u(t) + \Psi h(t) \in \mathcal{U}$.

Step 3. We prove that operator Ψ is continuous and $\Psi(\mathcal{U})$ is relatively compact.

For $h(t) \in \mathcal{U}$, taking a sequence $h_m(t)$, whose limit is $h(t)$. Obviously, when $t \in [-\tau + a, a]$, $h_m(t) \rightarrow h(t)$ means $\phi_m(t) \rightarrow \phi(t)$. Taking the norm of $\Psi h_m(t) - \Psi h(t)$, we obtain

$$\begin{aligned} & \|\Psi h_m(t) - \Psi h(t)\| \\ &= \sup_{a < t \leq T} \sum_{j=1}^n \left| \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \right. \\ & \quad \times \left(\sum_{k=1}^n a_{jk} f_k(h_{km}(s)) + \sum_{k=1}^n b_{jk} g_k(h_{km}(s - \tau)) \right. \\ & \quad \left. \left. - \sum_{k=1}^n a_{jk} f_k(h_k(s)) - \sum_{k=1}^n b_{jk} g_k(h_k(s - \tau)) \right) \right| \\ &\leq \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \\ & \quad \times \sum_{k=1}^n |a_{jk}| |f_k(h_{km}(s)) - f_k(h_k(s))| \\ & \quad + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \\ & \quad \times \sum_{k=1}^n |b_{jk}| |g_k(h_{km}(s - \tau)) - g_k(h_k(s - \tau))| \\ &\leq \sup_{a < t \leq T} \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\} \\ & \quad \times \sum_{k=1}^n |h_{km}(s) - h_k(s)| \\ & \quad + \sup_{a < t \leq T} \frac{1}{\Gamma(\alpha)} \sum_{s=a}^{\tau+a} (t - \rho(s))^{\overline{\alpha-1}} \sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\} \\ & \quad \times \sum_{k=1}^n |h_{km}(s - \tau) - h_k(s - \tau)| \\ & \quad + \sup_{a < t \leq T} \frac{1}{\Gamma(\alpha)} \sum_{s=\tau+a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\} \\ & \quad \times \sum_{k=1}^n |h_{km}(s - \tau) - h_k(s - \tau)| \\ &\leq \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \|h_m - h\| \\ & \quad + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \|\phi_m - \phi\|_C \\ & \quad + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \|h_m - h\|. \end{aligned} \quad (13)$$

Thus, we conclude from (13) that $\Psi h_m(t) \rightarrow \Psi h(t)$ as $h_m(t) \rightarrow h(t)$, which means that operator Ψ is continuous.

Next, we prove that $\Psi(\mathcal{U})$ is relatively compact. For $h(t) \in \mathcal{U}$, we can get from the proof of (12) that

$$\|\Psi h(t)\| \leq \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \|\phi\|_C + \frac{1}{\alpha B(T - a, \alpha)}$$

$$\begin{aligned} & \times \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N \\ & + \left(\frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \right. \\ & \left. + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \right) \eta \leq \eta, \end{aligned} \quad (14)$$

which means that operator Ψ is bounded.

Let $N_0 \in \mathbb{N}_a$, and $N_0 < t_1 < t_2$, we get from Definitions 2 and 4 that

$$\begin{aligned} & \|\Psi h(t_2) - \Psi h(t_1)\| \\ &= \sum_{j=1}^n \left| \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_2} (t_2 - \rho(s))^{\overline{\alpha-1}} \right. \\ & \quad \times \left(\sum_{k=1}^n a_{jk} f_k(h_k(s)) + \sum_{k=1}^n b_{jk} g_k(h_k(s - \tau)) \right) \\ & \quad \left. - \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_1} (t_1 - \rho(s))^{\overline{\alpha-1}} \right. \\ & \quad \times \left(\sum_{k=1}^n a_{jk} f_k(h_k(s)) + \sum_{k=1}^n b_{jk} g_k(h_k(s - \tau)) \right) \left. \right| \\ &= \sum_{j=1}^n \left| \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_1} \left((t_2 - \rho(s))^{\overline{\alpha-1}} - (t_1 - \rho(s))^{\overline{\alpha-1}} \right) \right. \\ & \quad \times \sum_{k=1}^n a_{jk} f_k(h_k(s)) \\ & \quad + \frac{1}{\Gamma(\alpha)} \sum_{s=t_1+1}^{t_2} (t_2 - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n a_{jk} f_k(h_k(s)) \\ & \quad + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_1} \left((t_2 - \rho(s))^{\overline{\alpha-1}} - (t_1 - \rho(s))^{\overline{\alpha-1}} \right) \\ & \quad \times \sum_{k=1}^n b_{jk} g_k(h_k(s - \tau)) \\ & \quad \left. + \frac{1}{\Gamma(\alpha)} \sum_{s=t_1+1}^{t_2} (t_2 - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n b_{jk} g_k(h_k(s - \tau)) \right| \\ &\leq \sum_{j=1}^n \left(\frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_1} \left| (t_2 - \rho(s))^{\overline{\alpha-1}} - (t_1 - \rho(s))^{\overline{\alpha-1}} \right| \right. \\ & \quad \times \sum_{k=1}^n |a_{jk}| |f_k(h_k(s))| \\ & \quad + \frac{1}{\Gamma(\alpha)} \sum_{s=t_1+1}^{t_2} (t_2 - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |a_{jk}| |f_k(h_k(s))| \left. \right) \\ & \quad + \sum_{j=1}^n \left(\frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_1} \left| (t_2 - \rho(s))^{\overline{\alpha-1}} - (t_1 - \rho(s))^{\overline{\alpha-1}} \right| \right. \\ & \quad \times \sum_{k=1}^n |b_{jk}| |g_k(h_k(s - \tau))| \\ & \quad + \frac{1}{\Gamma(\alpha)} \sum_{s=t_1+1}^{t_2} (t_2 - \rho(s))^{\overline{\alpha-1}} \sum_{k=1}^n |b_{jk}| |g_k(h_k(s - \tau))| \left. \right) \end{aligned}$$

$$\begin{aligned}
&\leq \sum_{j=1}^n \left(\sum_{k=1}^n |a_{jk}| M \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_1} \left((t_2 - \rho(s))^{\overline{\alpha-1}} - (t_1 - \rho(s))^{\overline{\alpha-1}} \right) \right. \\
&\quad \left. + \sum_{k=1}^n |a_{jk}| M \frac{1}{\Gamma(\alpha)} \sum_{s=t_1+1}^{t_2} (t_2 - \rho(s))^{\overline{\alpha-1}} \right) \\
&\quad + \sum_{j=1}^n \left(\sum_{k=1}^n |b_{jk}| N \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^{t_1} \left((t_2 - \rho(s))^{\overline{\alpha-1}} \right. \right. \\
&\quad \left. \left. - (t_1 - \rho(s))^{\overline{\alpha-1}} \right) + \sum_{k=1}^n |b_{jk}| N \frac{1}{\Gamma(\alpha)} \sum_{s=t_1+1}^{t_2} (t_2 - \rho(s))^{\overline{\alpha-1}} \right). \quad (15)
\end{aligned}$$

According to [Definitions 1](#) and [4](#), we obtain

$$\begin{aligned}
&\|\Psi \mathbf{h}(t_2) - \Psi \mathbf{h}(t_1)\| \\
&\leq \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M \left(\nabla_a^{-\alpha} (t_2 - a)^{\bar{0}} - \nabla_a^{-\alpha} (t_1 - a)^{\bar{0}} + \nabla_{t_1}^{-\alpha} (t_2 - t_1)^{\bar{0}} \right) \\
&\quad + \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N \left(\nabla_a^{-\alpha} (t_2 - a)^{\bar{0}} - \nabla_a^{-\alpha} (t_1 - a)^{\bar{0}} \right. \\
&\quad \left. + \nabla_{t_1}^{-\alpha} (t_2 - t_1)^{\bar{0}} \right) \\
&= \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M \frac{((t_2 - a)^{\bar{\alpha}} - (t_1 - a)^{\bar{\alpha}} + (t_2 - t_1)^{\bar{\alpha}})}{\Gamma(\alpha + 1)} \\
&\quad + \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N \frac{((t_2 - a)^{\bar{\alpha}} - (t_1 - a)^{\bar{\alpha}} + (t_2 - t_1)^{\bar{\alpha}})}{\Gamma(\alpha + 1)}, \quad (16)
\end{aligned}$$

which implies that $\|\Psi \mathbf{h}(t_2) - \Psi \mathbf{h}(t_1)\| \rightarrow 0$ as $t_2 \rightarrow t_1$. Thus, the operator Ψ is uniformly Cauchy in $\mathcal{U}$. According to [Lemma 4](#), the operator Ψ is relatively compact.

Step 4. We prove that there exists a solution $\mathbf{h}(t) \in \mathcal{U}$ when $\mathbf{h}(t) = \Phi \mathbf{h}(t) + \Psi \mathbf{h}(t)$.

In the discussion of the previous steps, it shows that $\Phi \mathbf{u}(t) + \Psi \mathbf{h}(t) \in \mathcal{U}$, for $\mathbf{u}(t), \mathbf{h}(t) \in \mathcal{U}$. If $\mathbf{h}(t) = \mathcal{E} \mathbf{h}(t) = \Phi \mathbf{h}(t) + \Psi \mathbf{h}(t)$, using Minkowski inequality, then one can get that

$$\begin{aligned}
\|\mathbf{h}(t)\| &= \|\Phi \mathbf{h}(t) + \Psi \mathbf{h}(t)\| \\
&= \sup_{a < t \leq T} \sum_{j=1}^n \left| \phi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t - \rho(s))^{\overline{\alpha-1}} \left(-c_j \mathbf{h}_j(s) \right. \right. \\
&\quad \left. \left. + W_j + \sum_{k=1}^n a_{jk} f_k(\mathbf{h}_k(s)) + \sum_{k=1}^n b_{jk} g_k(\mathbf{h}_k(s - \tau)) \right) \right| \\
&\leq \|\phi\|_C + \frac{1}{\alpha B(T - a, \alpha)} \|W\| + \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} \|\mathbf{h}\| \\
&\quad + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M + \frac{1}{\alpha B(T - a, \alpha)} \\
&\quad \times \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N \\
&\quad + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \|\mathbf{h}\| \\
&\quad + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \|\phi\|_C \\
&\quad + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \|\mathbf{h}\|
\end{aligned}$$

$$\begin{aligned}
&\leq \left(1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a + 1, \alpha)} \right) \|\phi\|_C + \frac{1}{\alpha B(T - a, \alpha)} \\
&\quad \times \|W\| + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M \\
&\quad + \frac{1}{\alpha B(T - a, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N + \left(\frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T - a, \alpha)} \right. \\
&\quad \left. + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T - a, \alpha)} \right. \\
&\quad \left. + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - a - \tau, \alpha)} \right) \eta \\
&= \wp \|\phi\|_C + \mathfrak{R} + \mathfrak{S} \eta \leq \eta, \quad (17)
\end{aligned}$$

which means that $\mathbf{h}(t) \in \mathcal{U}$. From [Lemma 5](#), the system (1) has a solution in $\mathcal{U}$. The proof is completed. $\square$

Remark 2. When the starting point $a = 0$, the system (1) turns into the following discrete fractional-order CVNNs:

$$\begin{cases} {}^c \nabla_0^\alpha \mathbf{h}_j(t) = -c_j \mathbf{h}_j(t) + \sum_{k=1}^n a_{jk} f_k(\mathbf{h}_k(t)) + \sum_{k=1}^n b_{jk} g_k \\ \quad \times (\mathbf{h}_k(t - \tau)) + W_j, \quad j = 1, 2, \dots, n, \quad t \in \mathbb{N}_0, \\ \mathbf{h}_j(s) = \phi_j(s), \quad s \in [-\tau, 0]. \end{cases} \quad (18)$$

The system (18) can be considered as a generalization of [Alzabut et al. \(2018\)](#) in the complex domain.

From the proof of [Theorem 1](#), we have the following result.

Corollary 1. Under [Assumptions 1](#) and [2](#), if the following conditions hold

$$\mathfrak{S} < 1, \quad (19)$$

$$\frac{\wp \|\phi\|_C + \mathfrak{R}}{1 - \mathfrak{S}} \leq \eta, \quad (20)$$

then the system (18) has a solution in $\mathcal{U} = \{\mathbf{h} : \mathbf{h} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n)^T, \|\mathbf{h}\| \leq \eta, \mathbf{h}_j \in L([0, T], \mathbb{C})\}$, where

$$\begin{aligned}
\mathfrak{S} &= \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T, \alpha)} \\
&\quad + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - \tau, \alpha)}, \\
\wp &= 1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T + 1, \alpha)}, \\
\mathfrak{R} &= \frac{1}{\alpha B(T, \alpha)} \|W\| + \frac{1}{\alpha B(T, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |a_{jk}| M \\
&\quad + \frac{1}{\alpha B(T, \alpha)} \sum_{j=1}^n \sum_{k=1}^n |b_{jk}| N.
\end{aligned}$$

3.2. Finite-time stability of solutions

Theorem 2. Under [Assumption 1](#), the system (1) is finite-time stable with respect to $(t_0 = a, \delta, \varepsilon, \Theta = [a, T])$, if there exist three positive constants T, δ, ε and satisfying $\delta < \varepsilon$, such that

$$\mathfrak{S} < 1, \quad (21)$$

$$\frac{\wp}{1 - \mathfrak{S}} < \frac{\varepsilon}{\delta}, \quad (22)$$

$$\text{where } \mathfrak{S} = \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T-a, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T-a, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a-\tau, \alpha)},$$

$$\wp = 1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a+1, \alpha)}.$$

Proof. Suppose there exist two solutions $\mathbf{h}(t) = (\mathbf{h}_1(t), \mathbf{h}_2(t), \dots, \mathbf{h}_n(t))^T$ and $\mathbf{u}(t) = (u_1(t), u_2(t), \dots, u_n(t))^T$ of system (1), whose initial conditions are given by ϕ and ψ . Based on Lemma 3, we can rewrite the $\mathbf{h}_j(t)$ and $u_j(t)$ in the following forms

$$\mathbf{h}_j(t) = \mathcal{E} \mathbf{h}_j(t) = \phi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t-\rho(s))^{\alpha-1} \left(-c_j \mathbf{h}_j(s) \right. \\ \left. + \sum_{k=1}^n a_{jk} f_k(\mathbf{h}_k(s)) + \sum_{k=1}^n b_{jk} g_k(\mathbf{h}_k(s-\tau)) + W_j \right),$$

$$u_j(t) = \mathcal{E} u_j(t) = \psi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t-\rho(s))^{\alpha-1} \left(-c_j u_j(s) \right. \\ \left. + \sum_{k=1}^n a_{jk} f_k(u_k(s)) + \sum_{k=1}^n b_{jk} g_k(u_k(s-\tau)) + W_j \right).$$

When $t = a$, then $\mathbf{h}_j(t) = \phi_j(a)$ and $u_j(t) = \psi_j(a)$. Taking the norm of $\|\mathbf{h}(t) - \mathbf{u}(t)\|$, one can obtain

$$\|\mathbf{h}(t) - \mathbf{u}(t)\| = \sup_{-\tau+a \leq s \leq a} \sum_{j=1}^n |\phi_j(a) - \psi_j(a)| \\ = \sum_{j=1}^n |\phi_j(a) - \psi_j(a)| = \|\phi(a) - \psi(a)\|_C \\ \leq \|\phi - \psi\|_C. \quad (23)$$

When $\|\phi - \psi\|_C \leq \delta$, we have that $\|\mathbf{h}(t) - \mathbf{u}(t)\| < \varepsilon$.

When $a < t \leq T$, under Assumption 1, Definitions 1 and 4, the solutions follow that

$$\|\mathbf{h}(t) - \mathbf{u}(t)\| \\ = \sup_{a < t \leq T} \sum_{j=1}^n \left| \phi_j(a) - \psi_j(a) + \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t-\rho(s))^{\alpha-1} \right. \\ \times \left(-c_j (\mathbf{h}_j(s) - u_j(s)) \right. \\ \left. + \sum_{k=1}^n a_{jk} (f_k(\mathbf{h}_k(s)) - f_k(u_k(s))) \right. \\ \left. + \sum_{k=1}^n b_{jk} (g_k(\mathbf{h}_k(s-\tau)) - g_k(u_k(s-\tau))) \right) \Big| \\ \leq \sum_{j=1}^n |\phi_j(a) - \psi_j(a)| + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \\ \times \sum_{s=a+1}^t (t-\rho(s))^{\alpha-1} c_j |\mathbf{h}_j(s) - u_j(s)| \\ + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t-\rho(s))^{\alpha-1} \\ \times \sum_{k=1}^n |a_{jk}| |f_k(\mathbf{h}_k(s)) - f_k(u_k(s))| \\ + \sup_{a < t \leq T} \sum_{j=1}^n \frac{1}{\Gamma(\alpha)} \sum_{s=a+1}^t (t-\rho(s))^{\alpha-1} \\ \times \sum_{k=1}^n |b_{jk}| |g_k(\mathbf{h}_k(s-\tau)) - g_k(u_k(s-\tau))|$$

$$\leq \|\phi - \psi\|_C + \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T-a, \alpha)} \|\mathbf{h} - \mathbf{u}\| \\ + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T-a, \alpha)} \|\mathbf{h} - \mathbf{u}\| \\ + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a+1, \alpha)} \|\phi - \psi\|_C \\ + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a-\tau, \alpha)} \|\mathbf{h} - \mathbf{u}\| \\ = \left(\frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T-a, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T-a, \alpha)} \right. \\ \left. + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a-\tau, \alpha)} \right) \|\mathbf{h} - \mathbf{u}\| \\ + \left(1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a+1, \alpha)} \right) \|\phi - \psi\|_C, \quad (24)$$

which is equivalent to

$$\|\mathbf{h} - \mathbf{u}\| \leq \frac{1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a+1, \alpha)}}{1 - \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T-a, \alpha)} - \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T-a, \alpha)} - \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-a-\tau, \alpha)}} \\ \times \|\phi - \psi\|_C = \frac{\wp}{1 - \mathfrak{S}} \|\phi - \psi\|_C. \quad (25)$$

When $\|\phi - \psi\|_C < \delta$, we have from conditions (21) and (22) that $\|\mathbf{h} - \mathbf{u}\| < \varepsilon$ for $t \in (a, T]$. Therefore, the system is finite-time stable on $(t_0 = a, \delta, \varepsilon, \Theta = [a, T])$ from Definition 3. The proof is completed. $\square$

When the starting point $a = 0$, from the proof of Theorem 2, we derive the following corollary.

Corollary 2. Under Assumption 1, the system (18) is finite-time stable with respect to $(t_0 = 0, \delta, \varepsilon, \Theta = [0, T])$, if there exist three positive constants T, δ, ε and satisfying $\delta < \varepsilon$, such that

$$\mathfrak{S} < 1, \quad (26)$$

$$\frac{\wp}{1 - \mathfrak{S}} < \frac{\varepsilon}{\delta}, \quad (27)$$

$$\text{where } \mathfrak{S} = \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T-\tau, \alpha)},$$

$$\wp = 1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T+1, \alpha)}.$$

Remark 3. From the conditions in Theorems 1 and 2, we can obtain the following conclusion: With the lower order, the discrete fractional-order CVNNs (1) will achieve the finite-time stability more easily. In fact, when the conditions of the theorems hold, we know from the properties of Beta function that $T - a - \tau > 0$. It can derive that $T - a > 1$ since τ is a positive integer. We note that $T \in \mathbb{N}_a$ and $\Gamma(s+1) = s\Gamma(s)$ for $s > 0$, it is easy to calculate that

$$\alpha B(T-a, \alpha) = \frac{\Gamma(T-a)}{(T-a+\alpha-1) \cdots (\alpha+2)(\alpha+1)},$$

$$\alpha B(T-a-\tau, \alpha) = \frac{\Gamma(T-a-\tau)}{(T-a-\tau+\alpha-1) \cdots (\alpha+2)(\alpha+1)},$$

$$\alpha B(T-a+1, \alpha) = \frac{\Gamma(T-a+1)}{(T-a+\alpha) \cdots (\alpha+2)(\alpha+1)}.$$

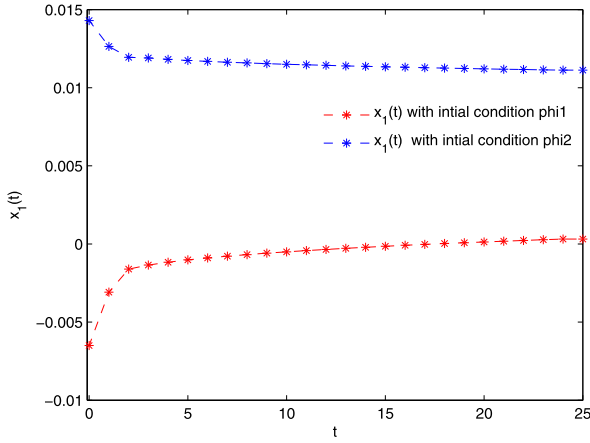


Fig. 1. State trajectories of real part $x_1(t)$ for system (28).

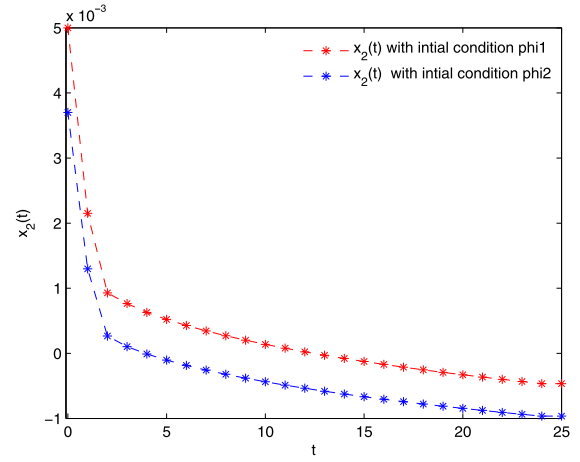


Fig. 2. State trajectories of real part $x_2(t)$ for system (28).

For given parameters $c_j, a_{jk}, b_{jk}, l_k, \lambda_k, T, \tau, \eta, \varepsilon$ and δ , the values of $\Im, \frac{\varphi \|\phi\|_C + \Re}{1 - \Im}$ and $\frac{\varphi}{1 - \Im}$ are decreasing with the decrease of order α , which means that the conditions (5), (6), (21) and (22) in Theorems 1 and 2 are more easily satisfied. Thus, the discrete fractional-order CVNNs (1) will achieve the finite-time stability more easily.

Remark 4. Many scholars have discussed the uniform stability, global asymptotic stability and finite-time stability of fractional-order CVNNs with time delays in Ding et al. (2017), Rakkiyappan et al. (2015, 2014), Wang et al. (2017a, 2017b) and Zhang, Song et al. (2017). Most of them considered that the activation functions of complex-valued systems can be separated into their real parts and imaginary parts. Thus they transformed CVNNs to equivalent RVNNs to analyze their dynamic behavior. However, this method will increase the dimension of systems and brings the difficulties on analysis. Compared with the existing literature, regardless of the activity functions are separable, the provided existence and finite-time stability criteria for discrete fractional-order CVNNs are valid and feasible in this paper.

Remark 5. In Abdeljawad (2011, 2013), Alzabut et al. (2018), Atici and Eloe (2009, 2012), Ganji and Gharari (2018) and Wu and Baleanu (2014), the authors studied the dynamics prosperities of discrete fractional difference equations in real field. However, there are very few results about discrete fractional-order system in complex field. Different from the existing literature, we first investigate discrete fractional-order CVNNs and analyze its dynamic behavior.

4. Examples

Example 1. Consider the following two neurons discrete fractional-order CVNNs with the starting point $a = 0$:

$${}^c \nabla_0^\alpha \tilde{h}_j(t) = -c_j \tilde{h}_j(t) + \sum_{k=1}^2 a_{jk} f_k(\tilde{h}_k(t)) + \sum_{k=1}^2 b_{jk} g_k(\tilde{h}_k(t - \tau)) + W_j, \quad j = 1, 2, \quad (28)$$

where $\alpha = 0.385$, $\tilde{h}_j(t) = x_j(t) + y_j(t)i$, $f_k(\tilde{h}_k) = 0.15 \tanh(x_k) + 0.15 \tanh(y_k)i$, $g_k(\tilde{h}_k) = 0.1 \tanh(x_k) + 0.1 \tanh(y_k)i$, $\tau = 1$, $k = 1, 2$, and

$$C = \begin{pmatrix} 0.1725 & 0 \\ 0 & 0.1884 \end{pmatrix},$$

$$A = \begin{pmatrix} -0.0262 - 0.0397i & 0.0525 - 0.0365i \\ 0.0352 + 0.0064i & -0.0576 + 0.0134i \end{pmatrix},$$

$$B = \begin{pmatrix} 0.0205 - 0.0755i & 0.1011 - 0.1011i \\ 0.0154 + 0.0250i & -0.0154 + 0.0506i \end{pmatrix},$$

$$W = \begin{pmatrix} 0.0013 - 0.0012i \\ -0.0011 + 0.0019i \end{pmatrix}.$$

$$\text{The initial condition is } \phi = \begin{pmatrix} -0.0065 + 0.0013i \\ 0.005 \end{pmatrix}.$$

The activation functions satisfy Assumptions 1 and 2 with $l_k = 0.15$, $\lambda_k = 0.1$, $M = 0.15\sqrt{2}$, $N = 0.1\sqrt{2}$, $k = 1, 2$. Take $\delta = 0.052$, $\varepsilon = 0.5$, from the conditions (19) and (27), we have

$$\frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - \tau, \alpha)} < 1,$$

$$\frac{1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T + 1, \alpha)}}{1 - \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T, \alpha)} - \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T, \alpha)} - \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - \tau, \alpha)}} < \frac{\varepsilon}{\delta} = 9.6154.$$

Then it can estimate the time $T \approx 25$ and derive that $\varphi = 1.0869$, $\Re = 0.3508$, $\Im = 0.878 < 1$, $\|\tilde{h}\| = 0.0074 < \eta = 2.9784$, $\frac{\varphi}{1 - \Im} = 8.9062 < 9.6154$. Obviously, conditions (19), (20) and (27) hold. From Theorems 1 and 2, the system (28) has a solution and is finite-time stable on $(t_0 = 0, \delta = 0.052, \varepsilon = 0.5, \Theta = [0, 25])$. Figs. 1–4 display the time response trajectories of system (28) with two initial values $\phi = \begin{pmatrix} -0.0065 + 0.0013i \\ 0.005 \end{pmatrix}$

and $\psi = \begin{pmatrix} 0.0143 + 0.0033i \\ 0.0037 - 0.0093i \end{pmatrix}$, where x_j denotes the real part of $\tilde{h}_j$, y_j denotes the imaginary part of $\tilde{h}_j$, ϕ and ψ denote the initial conditions ϕ and ψ , $j = 1, 2$. It can be seen that when the initial error does not exceed δ , the difference between any two trajectories of the system (28) during the interval time $[0, 25]$ does not exceed ε . Fig. 5 shows the exact error between the two solutions for the system, where $\|e(t)\| = \|\tilde{h}(t, 0, \phi) - \tilde{h}(t, 0, \psi)\| = 0.4839 < 0.5$. Therefore, the results of Theorem 2 are verified by employing this simulation.

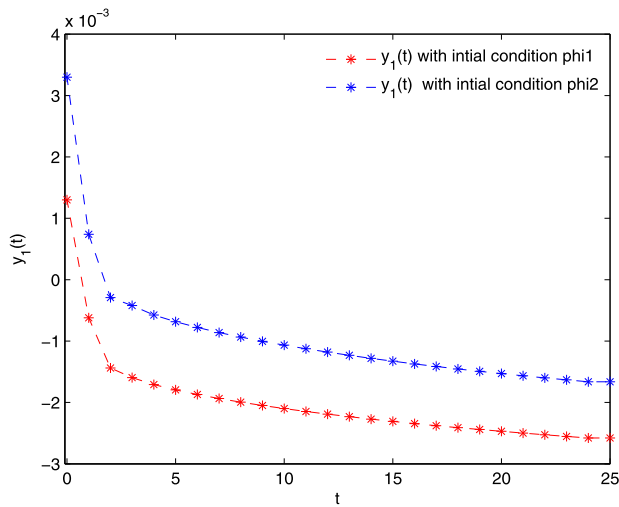


Fig. 3. State trajectories of imaginary part $y_1(t)$ for system (28).

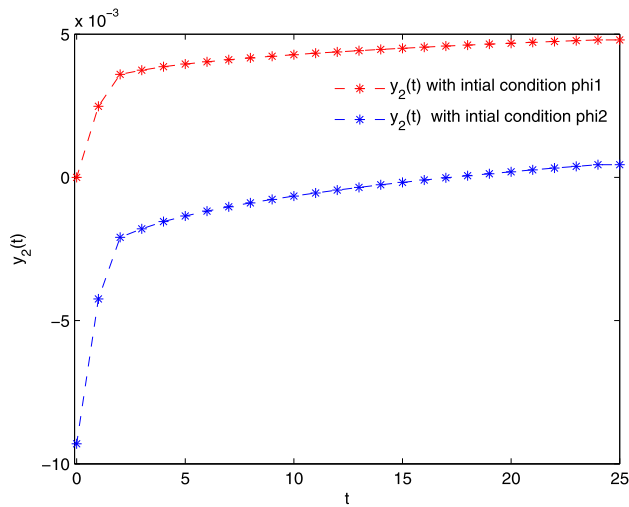


Fig. 4. State trajectories of imaginary part $y_2(t)$ for system (28).

Example 2. Consider the following three neurons discrete fractional-order CVNNs with the starting point $a = 0$:

$${}^c \nabla_0^\alpha \tilde{h}_j(t) = -c_j \tilde{h}_j(t) + \sum_{k=1}^3 a_{jk} f_k(\tilde{h}_k(t)) + \sum_{k=1}^3 b_{jk} g_k(\tilde{h}_k(t - \tau)) + W_j, \quad j = 1, 2, 3, \quad (29)$$

where $\alpha = 0.48$, $\tilde{h}_j(t) = x_j(t) + y_j(t)i$, $f_k(\tilde{h}_k) = 0.03(|x_k + 1| - |x_k - 1|) + 0.03(|y_k + 1| - |y_k - 1|)i$, $g_k(\tilde{h}_k) = 0.03(|x_k + 1| - |x_k - 1|) + 0.03(|y_k + 1| - |y_k - 1|)i$, $\tau = 2$, $k = 1, 2, 3$, and

$$C = \begin{pmatrix} 0.1908 & 0 & 0 \\ 0 & 0.1827 & 0 \\ 0 & 0 & 0.1964 \end{pmatrix},$$

$$A = \begin{pmatrix} 0.08 - 0.06i & 0.078 - 0.05i & 0.045 + 0.07i \\ -0.06 - 0.05i & 0.045 + 0.07i & 0.058 + 0.06i \\ -0.072 + 0.054i & 0.046 + 0.048i & 0.012 - 0.041i \end{pmatrix},$$

$$B = \begin{pmatrix} 0.042 + 0.053i & -0.052 + 0.064i & 0.045 + 0.06i \\ -0.084 + 0.056i & -0.047 - 0.032i & 0.048 - 0.037i \\ 0.048 - 0.047i & 0.045 + 0.052i & 0.051 - 0.062i \end{pmatrix},$$

$$W = \begin{pmatrix} -0.0352 - 0.0339i \\ 0.0171 - 0.0248i \\ -0.0264 - 0.0367i \end{pmatrix}.$$

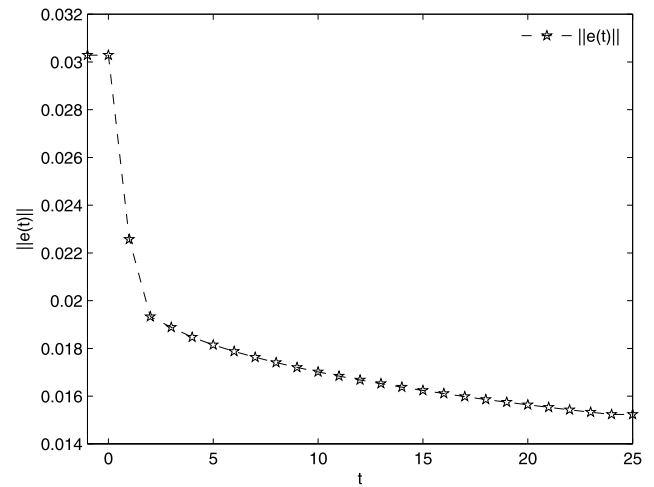


Fig. 5. The time evolution trajectory of error norm of solutions for system (28).

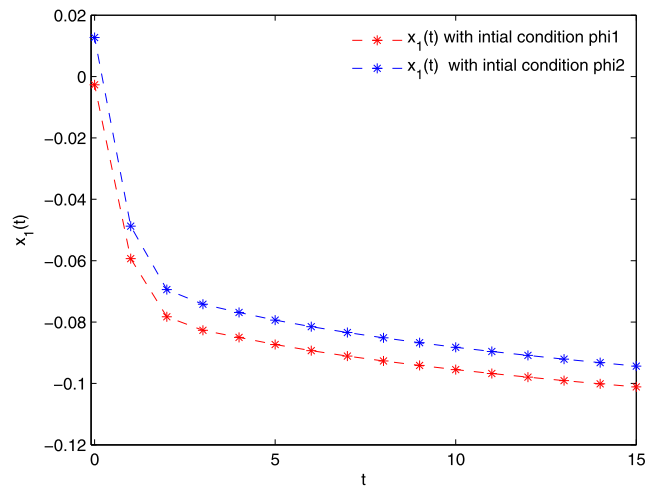


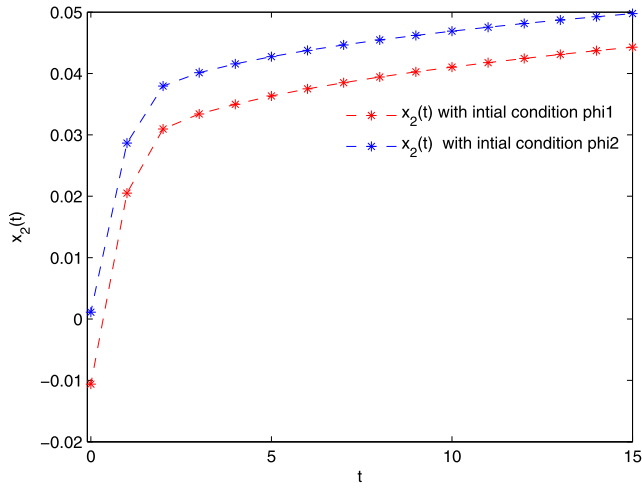
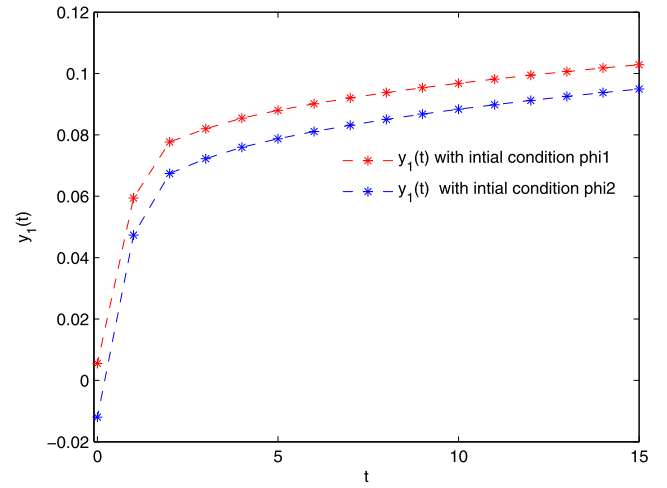
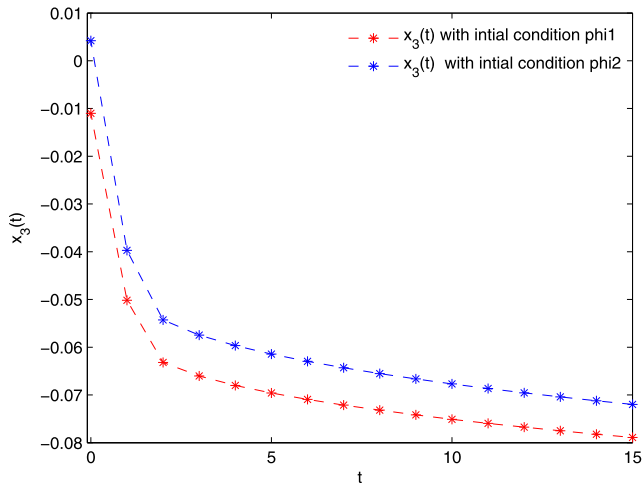
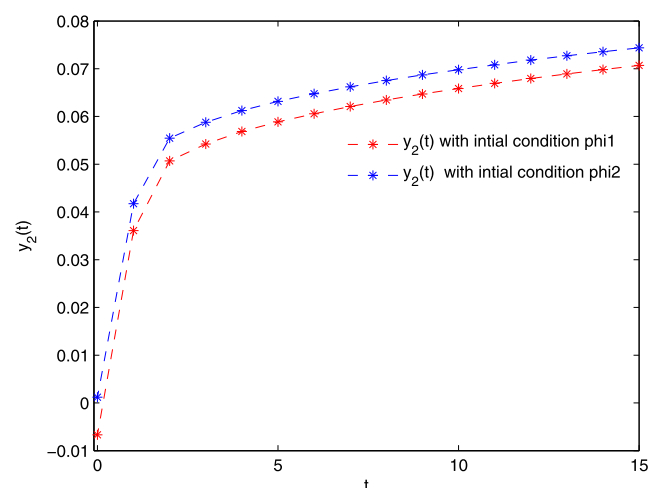
Fig. 6. State trajectories of real part $x_1(t)$ for system (29).

The initial condition is $\phi = \begin{pmatrix} -0.0027 + 0.0055i \\ -0.0106 - 0.0067i \\ -0.011 - 0.009i \end{pmatrix}$.

The activation functions satisfy [Assumptions 1](#) and [2](#) with $l_k = 0.06$, $\lambda_k = 0.06$, $M = 0.06\sqrt{2}$, $N = 0.06\sqrt{2}$, $k = 1, 2, 3$. Take $\delta = 0.056$, $\varepsilon = 1$, from the conditions [\(19\)](#) and [\(27\)](#), we have

$$\begin{aligned} & \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T, \alpha)} + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T, \alpha)} \\ & + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - \tau, \alpha)} < 1, \\ & 1 + \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T + 1, \alpha)} \\ & 1 - \frac{\max_{1 \leq j \leq n} \{c_j\}}{\alpha B(T, \alpha)} - \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|a_{jk}| l_k\}}{\alpha B(T, \alpha)} - \frac{\sum_{j=1}^n \max_{1 \leq k \leq n} \{|b_{jk}| \lambda_k\}}{\alpha B(T - \tau, \alpha)} \\ & < \frac{\varepsilon}{\delta} = 17.8571. \end{aligned}$$

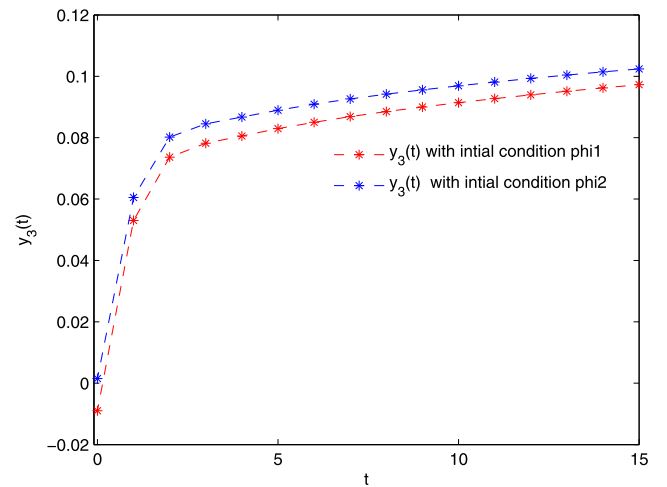
Then it can estimate the time $T \approx 15$ and derive that $\wp = 1.0671$, $\Re = 0.991$, $\Im = 0.9354 < 1$, $\|\tilde{h}\| = 0.3529 < \eta = 15.8849$, $\frac{\wp}{1 - \Im} = 16.5188 < 17.8571$. Obviously, conditions [\(19\)](#), [\(20\)](#) and [\(27\)](#) hold. From [Theorems 1](#) and [2](#), the system [\(29\)](#) has a solution and is finite-time stable on $(t_0 = 0, \delta = 0.056, \varepsilon = 1, \Theta = [0, 15])$. [Figs. 6–12](#) display the time responses of system

Fig. 7. State trajectories of real part $x_2(t)$ for system (29).Fig. 9. State trajectories of imaginary part $y_1(t)$ for system (29).Fig. 8. State trajectories of real part $x_3(t)$ for system (29).Fig. 10. State trajectories of imaginary part $y_2(t)$ for system (29).

(29) with two initial values $\phi = \begin{pmatrix} -0.0027 + 0.0055i \\ -0.0106 - 0.0067i \\ -0.011 - 0.009i \end{pmatrix}$ and $\psi = \begin{pmatrix} 0.0127 - 0.0120i \\ 0.0011 + 0.0012i \\ 0.0042 + 0.0015i \end{pmatrix}$, where x_j denotes the real part of h_j , y_j denotes the imaginary part of h_j , ϕ and ψ denote the initial conditions ϕ and ψ , $j = 1, 2, 3$. It can be seen that when the initial error does not exceed δ , the difference between any two trajectories of the system (29) during the interval time $[0, 15]$ does not exceed ε . Fig. 12 shows the exact error between the two solutions for the system, where $\|e(t)\| = \|\hat{h}(t, 0, \phi) - u(t, 0, \psi)\| = 0.6057 < 1$. Therefore, the results of Theorem 2 are verified by means of this simulation.

5. Conclusion

In this paper, the existence and finite-time stability of discrete fractional-order CVNNs with time delays have been investigated. Based on the theory of discrete fractional calculus, we propose a new discrete Caputo fractional difference equation and research its dynamic properties. Afterwards, by making use of Arzela-Ascoli's theorem, inequality scaling skills and fixed point theorem, several sufficient criteria of delay-dependent have been derived to guarantee the existence and finite-time stability of

Fig. 11. State trajectories of imaginary part $y_3(t)$ for system (29).

solutions for the proposed networks. Finally, numerical examples are also presented to manifest the feasibility and validity of the obtained results. Furthermore, we have drawn a conclusion: with the lower order, the discrete fractional-order complex-valued

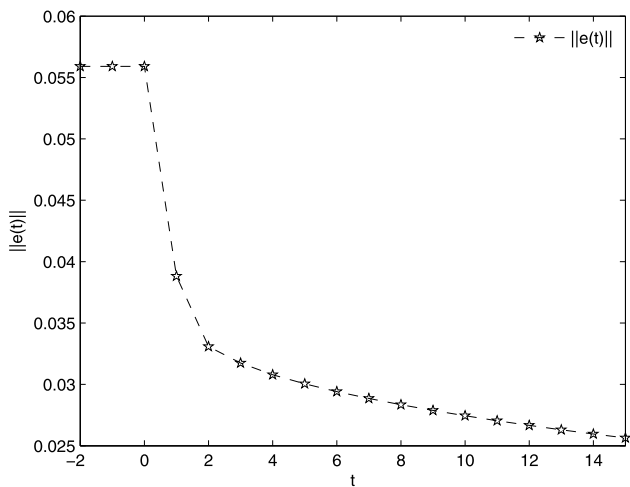


Fig. 12. The time evolution trajectory of error norm of solutions for system (29).

neural networks (1) will achieve the finite-time stability more easily. We would like to point out that it is possible to extend our approach to achieve other fractional-order models, such as discrete fractional-order neural networks with time-varying delays, which may be a good topic. The corresponding results will be carried out in the near future.

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